Mean-field limit of the carrier game: statements and proofs

Quang-Vinh Dang

Working notes (the same text is input by mfg/paper/main.tex)

1 Model

1.1 The time-expanded zone graph

The city is divided into zones $z \in [Z]$ and the delivery day into periods $t \in [H]$ (in the experiments $H = 10$ hourly periods from 08:00 to 18:00). The *time-expanded zone graph* G_H has a node (z, t) for every zone and period, a travel arc $(z, t) \to (z', t')$ whenever a vehicle that leaves z in period t reaches z' in period $t' \geq t$ (travel times depend on the departure period), and a waiting arc $(z, t) \to (z, t + 1)$. Because every arc moves forward in time, G_H is acyclic.

A *tour* of a carrier is a path in G_H that starts and ends at the carrier's depot and makes deliveries at some of the nodes it visits. Delivering in zone z during period t adds to the deliveries q_{zt} of the carrier at that node. The coordinates of the interaction are the nodes $k = (z, t)$ touched by some tour; we write $\mathcal{K}$ for this set and $d = |\mathcal{K}| \leq ZH$. Vehicle-hours spent in zone z during period t are $r\, q_{zt}$, where r is the in-zone time per delivery (assumption C3 of 3): congestion is caused by the time spent serving a zone, and inter-zone driving is priced through the tour's own cost only. This is the delivery-unit reduction of the Omega paper.

Demand and congestion In coordinate k, the inverse demand for deliveries is $\alpha_k - b_k Q_k$ with $b_k > 0$, where Q_k is the total number of deliveries made at k by all carriers. A vehicle-hour at k costs $\gamma(\xi_k + c_k X_k)$ where $X_k = rQ_k$ is the congestion variable. Per delivery, a carrier therefore pays $\gamma r \xi_k$ (part of its private cost) plus $\kappa_k Q_k$ with $\kappa_k = \gamma c_k r^2 > 0$. Write

$$m_k = b_k + \kappa_k, \qquad M = \operatorname{diag}(m), \qquad \varrho = \max_k \kappa_k / m_k < 1.$$

For $v \in \mathbb{R}^d$ and a positive weight vector w we write $\|v\|_w^2 = \sum_k w_k v_k^2$; thus $\|v\|_m$ is the M-norm and $\|v\|_{m^{-1}}$ its dual norm.

1.2 Carriers, plans and their convexification

There are N carriers. Carrier i has a finite set $\mathcal{P}_i$ of *plans*: a plan is a multiset of tours that the carrier's fleet can operate in one day. Plan p has deliveries $q_i(p) \in \mathbb{R}^d_{\geq 0}$ and private profit $\pi_i(p) = \sum_k \alpha_k q_{ik}(p) - C_i(p)$, where $C_i(p)$ is its routing cost (driver time, fuel, fixed costs and the free-flow congestion term $\gamma r \langle \xi, q_i(p) \rangle$). A *small multi-stop carrier* is a carrier whose fleet is small compared with the market; the mean-field limit lets the share of every carrier go to zero while plans keep their routing structure.

For most results the plan set is convexified. A carrier with a fleet budget B_i (vehicle-hours) and a tour set $\mathcal{T}_i$ chooses fleet weights $y_i \in Y_i = \{y \in \mathbb{R}^{\mathcal{T}_i}_{\geq 0} : \langle w_i, y \rangle \leq B_i\}$, where w_{ij} is the duration of

tour j; its deliveries are $a_i = L_i y_i$ with $L_i \geq 0$ entrywise (column j is the delivery vector of tour j per vehicle-hour) and its profit is $\pi_i(y_i) = \langle p_i, y_i \rangle$. The abstract assumptions under which the theory holds are:

(A1) Y_i is a nonempty compact convex subset of a Euclidean space, $\pi_i : Y_i \to \mathbb{R}$ is concave and upper semicontinuous, and $a_i = L_i y_i$ with L_i linear and $L_i y_i \in \mathbb{R}^d_{\geq 0}$ for $y_i \in Y_i$;

(A2) $m_k > 0$ for all k.

The fleet polytope with linear profits satisfies (A1). We write $A = \sum_i a_i$ (in delivery units $A = Q$) and

$$\bar{s} = \max_i \max_{y \in Y_i} \max_k (L_i y)_k$$

for the largest number of deliveries that one carrier can make at one coordinate; $\bar{s}$ is the *share* parameter of the limit.

Payoffs and equilibria The city posts a charge $u \in \mathbb{R}^d$ per delivery. Carrier i's payoff is

$$J_i(y_i; y_{-i}) = \pi_i(y_i) - \langle a_i, M(A_{-i} + a_i) \rangle - \langle u, a_i \rangle, \qquad A_{-i} = \sum_{j \neq i} a_j. \tag{1}$$

A *Nash equilibrium* y^{N} is a profile in which each y_i maximises $J_i(\cdot\,; y_{-i})$: a carrier takes its own effect on the price and the congestion of the coordinates it serves into account (Cournot behaviour). A *Wardrop* (*mean-field*) *equilibrium* y^{W} is a profile in which each y_i maximises $\pi_i(y_i) - \langle a_i, MA + u \rangle$ with the aggregate $A = A(y^{\mathrm{W}})$ held fixed: carriers are price takers. The mean-field reading is the usual one: if carrier i is replaced by a continuum of identical infinitesimal agents whose plans sum to a point of Y_i, each agent's own effect vanishes and the equilibrium conditions of the continuum are those of y^{W}. On G_H this is the Wardrop equilibrium of a Markov-decision-process congestion game in which vehicles choose paths [2], with an elastic (Cournot) price added; it is also a discrete-time, finite-state potential mean-field game [4, 5].

The welfare of a profile, with charges treated as transfers, is

$$\mathrm{SW}(y) = \sum_i \pi_i(y_i) + \sum_k \Big[-\tfrac{b_k}{2} Q_k^2 - \kappa_k Q_k^2 \Big] = \sum_i \pi_i(y_i) - \tfrac{1}{2} \|Q\|^2_{m+\kappa}, \tag{2}$$

the gross consumer benefit $\alpha_k Q_k - \frac{b_k}{2} Q_k^2$ (its linear part is inside π_i) minus routing and congestion costs. A *social optimum* maximises SW.

1.3 Potentials and a linearisation lemma

The first lemma is elementary convex analysis; it is used in almost every proof below.

Lemma 1 (Linearisation)**.** *Let C be convex, $h : C \to \mathbb{R}$ concave and g convex and differentiable on a neighbourhood of C. If x^* maximises $h - g$ over C, then x^* maximises $h - \langle \nabla g(x^*), \cdot \rangle$ over C.*

Proof. Let $x \in C$ and $x_t = x^* + t(x - x^*)$ for $t \in (0, 1]$. Optimality and concavity of h give $0 \geq h(x_t) - g(x_t) - h(x^*) + g(x^*) \geq t\,(h(x) - h(x^*)) - (g(x_t) - g(x^*))$. Divide by t and let $t \downarrow 0$: $0 \geq h(x) - h(x^*) - \langle \nabla g(x^*), x - x^* \rangle$. □

Lemma 2 (Potentials, uniqueness and the own surcharge)**.** *Assume (A1)–(A2) and fix u. Let*

$$\Phi_{\mathrm{N}}(y) = \sum_i \pi_i(y_i) - \langle u, A \rangle - \tfrac{1}{2} \|A\|^2_m - \tfrac{1}{2} \sum_i \|a_i\|^2_m, \qquad \Phi_{\mathrm{W}}(y) = \sum_i \pi_i(y_i) - \langle u, A \rangle - \tfrac{1}{2} \|A\|^2_m.$$

(a) y is a Nash equilibrium iff it maximises Φ_{N} over $\prod_i Y_i$, and a Wardrop equilibrium iff it maximises Φ_{W}; a social optimum maximises Φ_{W} with $u = 0$ and m replaced by $m + \kappa$. All three sets are nonempty.

(b) The Nash contributions a_i^{N} are unique, and so is the Wardrop aggregate A^{W}; the Wardrop contributions need not be.

(c) (Own surcharge.) Every Nash equilibrium is a Wardrop equilibrium of the game in which carrier i pays the additional per-delivery charge $\sigma_i = Ma_i^{\mathrm{N}}$ (with a_i^{N} held fixed): each y_i^{N} maximises $\pi_i(y_i) - \langle a_i, MA^{\mathrm{N}} + u + Ma_i^{\mathrm{N}} \rangle$.

Proof. (a) $\Phi_{\mathrm{N}} - J_i$ does not depend on y_i: the terms of Φ_{N} containing y_i are $\pi_i - \langle u, a_i \rangle - \langle A_{-i}, Ma_i \rangle - \|a_i\|_m^2$, which are those of J_i. Hence a maximiser of Φ_{N} is a Nash equilibrium. Conversely let y be a Nash equilibrium and write $\Phi_{\mathrm{N}} = \sum_i \pi_i - f$ with f convex and differentiable. By Lemma 1 applied to carrier i (with $h = \pi_i$, $g = f(\cdot, y_{-i})$), $\pi_i(y_i') - \langle \nabla_i f(y), y_i' - y_i \rangle \le \pi_i(y_i)$ for all $y_i' \in Y_i$. Summing over i and using $f(y') \ge f(y) + \langle \nabla f(y), y' - y \rangle$ gives $\Phi_{\mathrm{N}}(y') \le \Phi_{\mathrm{N}}(y)$. The Wardrop case is the same with the linear functional $y_i \mapsto \langle a_i, MA + u \rangle$, whose y_i-gradient is $\nabla_i(\frac{1}{2}\|A\|_m^2 + \langle u, A \rangle)$. Existence: the potentials are upper semicontinuous on a compact set. (b) Φ_{N} is the sum of a concave function and $-\frac{1}{2}\|A\|_m^2 - \frac{1}{2}\sum_i \|a_i\|_m^2$, which is strictly concave in $(a_1, \dots, a_N)$; two maximisers with different contributions would have a midpoint with a strictly larger value. The same argument with $-\frac{1}{2}\|A\|_m^2$ gives uniqueness of A^{W}. Non-uniqueness of Wardrop contributions: two identical carriers with linear profits can exchange deliveries. (c) Lemma 1 applied to J_i with $g(y_i) = \|L_i y_i\|_m^2$, whose gradient at y_i^{N} is $2L_i^\top M a_i^{\mathrm{N}}$, and $A_{-i} + a_i^{\mathrm{N}} = A^{\mathrm{N}}$. □

Parts (a) and (b) are classical: the Nash potential of a Cournot game with linear demand is due to Slade [9] and the Wardrop potential to Beckmann et al. [1]; for MDP congestion games see Calderone and Sastry [2]. Part (c) is the device behind all the bounds below: the finite game is the mean-field game in which each carrier pays a private surcharge equal to its own marginal effect Ma_i^{N}, a vector of size at most $\max_k m_k \cdot \bar{s}$.

Replicas A *base type* θ is a triple $(\mathcal{Y}_\theta, \pi_\theta, L_\theta)$ satisfying (A1) with π_θ positively homogeneous (linear profits are). Its *n-split* consists of n carriers with $Y = \mathcal{Y}_\theta / n$, profit π_θ and the same L_θ: the fleet of the type is divided equally. A game whose carriers are the n_θ-splits of base types $\theta \in \Theta$ is a *replica game*.

Lemma 3 (Symmetry of replicas)**.** *In a replica game, carriers of the same type have the same Nash contributions, and the type totals $Y_\theta = \sum_{i \in \theta} y_i$ of a Nash equilibrium maximise*

$$\Phi^{(n)}(Y) = \sum_\theta \pi_\theta(Y_\theta) - \langle u, A \rangle - \tfrac{1}{2}\|A\|_m^2 - \tfrac{1}{2}\sum_\theta \frac{1}{n_\theta}\|L_\theta Y_\theta\|_m^2, \qquad A = \sum_\theta L_\theta Y_\theta,$$

over $\prod_\theta \mathcal{Y}_\theta$. The Wardrop aggregate does not depend on (n_θ).

Proof. Φ_{N} is invariant under permutations of identical carriers, so by uniqueness of the Nash contributions (Lemma 2(b)) identical carriers have equal contributions. Averaging a maximiser over the permutations within each type gives a symmetric maximiser (concavity), whose value is $\Phi^{(n)}$ of its type totals; conversely every Y yields the symmetric profile $y_i = Y_\theta / n_\theta$. For the Wardrop part, Φ_{W} depends on y only through the type totals, and $\sum_{i \in \theta} \mathcal{Y}_\theta / n_\theta = \mathcal{Y}_\theta$ by convexity. □

Lemma 3 is also the computational reduction used in Section : the Nash equilibrium of n replicas per type costs one convex quadratic program in the type totals, whatever n is.

1.4 Imprecise beliefs and vehicle counts

Two extensions of the Omega model are kept. Under Γ-maximin beliefs of width δ about the rival aggregate (Omega, Lemma "Γ-maximin surcharge") each carrier evaluates its plans at the upper corner $A_{-i}+\delta w$ of a box of weights $w \in \mathbb{R}^d_{>0}$; Section 4 studies this layer in the limit. Under vehicle-count congestion (D2) the congestion variable counts vehicles instead of vehicle-hours; Section 5 treats it.

2 How far is the finite market from its mean-field limit?

Throughout this section the charge u is fixed (all bounds hold uniformly in u), y^{N} is a Nash equilibrium and y^{W} a Wardrop equilibrium of the same carriers, and $\Delta A = A^{\mathrm{N}} - A^{\mathrm{W}}$.

2.1 A general bound and its tightness

Theorem 4 (General bound)**.** *Assume (A1)–(A2).*

(a) For every Wardrop equilibrium y^{W},

$$\|\Delta A\|_m^2 \le \sum_i \langle a_i^{\mathrm{N}}, M(a_i^{\mathrm{W}} - a_i^{\mathrm{N}})\rangle \le \frac{1}{4}\sum_i \|a_i^{\mathrm{W}}\|_m^2, \qquad \textit{hence} \qquad \|\Delta A\|_m \le \frac{1}{2}\Big(\sum_i \|a_i^{\mathrm{W}}\|_m^2\Big)^{1/2} \le \frac{1}{2}\big(\bar{s}\,\langle m, A^{\mathrm{W}}\rangle\big)$$

(b) (Lumpy plans.) Let p be a pure Nash equilibrium of the game in which carrier i chooses a plan in the finite set $\mathcal{P}_i$ (payoff (1) with $a_i = q_i(p_i)$), and let x^{W} be a Wardrop equilibrium of its mixed extension (carrier i mixes over $\mathcal{P}_i$, contributes the mean $\bar{a}_i$). Then

$$\|A(p) - A^{\mathrm{W}}\|_m^2 \le \sum_i \Big(\mathbb{E}_{x_i^{\mathrm{W}}}\|q_i\|_m^2 - \langle \bar{a}_i^{\mathrm{W}}, Mq_i(p_i)\rangle\Big) \le \bar{s}\,\langle m, A^{\mathrm{W}}\rangle,$$

with $\bar{s}$ the largest delivery of a plan at a coordinate. A pure Nash equilibrium exists.

Proof. (a) By Lemma 2(c), y_i^{N} maximises $\pi_i - \langle a_i, MA^{\mathrm{N}} + u + Ma_i^{\mathrm{N}}\rangle$, so it is at least as good as y_i^{W}; and y_i^{W} maximises $\pi_i - \langle a_i, MA^{\mathrm{W}} + u\rangle$, so it is at least as good as y_i^{N}:

$$\pi_i(y_i^{\mathrm{W}}) - \langle a_i^{\mathrm{W}}, MA^{\mathrm{N}} + u + Ma_i^{\mathrm{N}}\rangle \le \pi_i(y_i^{\mathrm{N}}) - \langle a_i^{\mathrm{N}}, MA^{\mathrm{N}} + u + Ma_i^{\mathrm{N}}\rangle,$$

$$\pi_i(y_i^{\mathrm{N}}) - \langle a_i^{\mathrm{N}}, MA^{\mathrm{W}} + u\rangle \le \pi_i(y_i^{\mathrm{W}}) - \langle a_i^{\mathrm{W}}, MA^{\mathrm{W}} + u\rangle.$$

Adding, the profits and charges cancel and $\langle a_i^{\mathrm{W}} - a_i^{\mathrm{N}}, M(A^{\mathrm{W}} - A^{\mathrm{N}})\rangle \le \langle a_i^{\mathrm{W}} - a_i^{\mathrm{N}}, Ma_i^{\mathrm{N}}\rangle$. Summing over i gives the first inequality. The second is $\langle x, M(y-x)\rangle \le \frac{1}{4}\|y\|_m^2$, i.e. $\|x - y/2\|_m^2 \ge 0$, with $x = a_i^{\mathrm{N}}$, $y = a_i^{\mathrm{W}}$. Finally $\sum_i \|a_i^{\mathrm{W}}\|_m^2 = \sum_k m_k \sum_i (a_{ik}^{\mathrm{W}})^2 \le \bar{s}\sum_k m_k A_k^{\mathrm{W}}$ because $0 \le a_{ik}^{\mathrm{W}} \le \bar{s}$.

(b) A pure equilibrium exists because Φ_{N} restricted to pure profiles is an exact potential of a finite game [6]. At p, carrier i does not gain by deviating to any plan p'; averaging over $p' \sim x_i^{\mathrm{W}}$ (payoffs are evaluated against the realised rival aggregate $A_{-i} = A(p) - q_i(p_i)$),

$$\pi_i(p_i) - \langle q_i(p_i), M(A_{-i} + q_i(p_i))\rangle \ge \bar{\pi}_i(x_i^{\mathrm{W}}) - \langle \bar{a}_i^{\mathrm{W}}, MA_{-i}\rangle - \mathbb{E}_{x_i^{\mathrm{W}}}\|q_i\|_m^2.$$

The Wardrop condition, tested against the pure strategies p, gives $\sum_i \bar{\pi}_i(x_i^{\mathrm{W}}) - \langle A^{\mathrm{W}}, MA^{\mathrm{W}}\rangle \ge \sum_i \pi_i(p_i) - \langle A(p), MA^{\mathrm{W}}\rangle$ (the charge terms are omitted; they cancel as in (a)). Adding the sum over i of the first inequality to the second and expanding A_{-i} gives $0 \ge \|A(p) - A^{\mathrm{W}}\|_m^2 + \sum_i \langle \bar{a}_i^{\mathrm{W}}, Mq_i(p_i)\rangle - \sum_i \mathbb{E}_{x_i^{\mathrm{W}}}\|q_i\|_m^2$, which is the claim; the last bound uses $\langle \bar{a}_i^{\mathrm{W}}, Mq_i(p_i)\rangle \ge 0$ and $\mathbb{E}\|q_i\|_m^2 \le \bar{s}\,\langle m, \bar{a}_i^{\mathrm{W}}\rangle$. □

The first inequality of (a) is the summed variational-inequality argument of Paccagnan et al. [7, Thm. 1(3)], written for this game; with $\bar{s} \sim 1/N$ it gives the rate $O(N^{-1/2})$ of that theorem. The constant $\frac{1}{2}$ and part (b) are additions. The next result shows that, without further structure, nothing better than $O(\bar{s}^{1/2})$ holds: the constant $\frac{1}{2}$ is attained up to the factor $1/\sqrt{2}$.

Proposition 5 (Tightness)**.** *Let $n \geq 2$ and $\varphi_n \in (0, \pi/2)$ with $1 - \cos\varphi_n = 1/n$; put $c = \cos\varphi_n$, $v = (c, \sin\varphi_n)$. Consider two coordinates with $M = I$ and $2n$ carriers: n carriers of type A choose $t \in [0, 1/n]$, deliver te_1 and earn $\alpha_A t$; n carriers of type B choose $t \in [0, 1/n]$, deliver tv and earn $\alpha_B t$; $(\alpha_A, \alpha_B) = (1, c)$. Then the Wardrop aggregate is $A^{\mathrm{W}} = e_1$ (type totals $T^{\mathrm{W}} = (1, 0)$), the Nash type totals are $T^{\mathrm{N}} = \big(\frac{3}{4} - \frac{1}{4n}, \frac{1}{4} - \frac{1}{4n}\big)$, and*

$$\|A^{\mathrm{N}} - A^{\mathrm{W}}\|_2^2 = \frac{1}{8n} + \frac{2n-1}{8n^3}, \qquad \text{so} \qquad \|\Delta A\|_2 \geq \frac{1}{2\sqrt{2n}} = \frac{1}{\sqrt{2}} \cdot \frac{1}{2}\Big(\sum_i \|a_i^{\mathrm{W}}\|^2\Big)^{1/2}.$$

Every carrier can deliver at most $\bar{s} = 1/n$ at any coordinate, while $\|\Delta A\|_\infty \geq \|\Delta A\|_2/\sqrt{2} \geq \frac{1}{4}\sqrt{n}\,\bar{s}$. Hence there is no constant C, independent of the instance, with $\|A^{\mathrm{N}} - A^{\mathrm{W}}\| \leq C\bar{s}$, even with two coordinates, linear profits and bounded data; over this class the worst-case gap is of exact order $\bar{s}^{1/2}$.

Proof. Write $J = [e_1, v]$ and $G = J^\top J = \left(\begin{smallmatrix} 1 & c \\ c & 1 \end{smallmatrix}\right) \succ 0$. By Lemma 3 it suffices to work with the type totals $T \in [0,1]^2$, $A = JT$. The Wardrop potential is $\langle \alpha, T\rangle - \frac{1}{2}T^\top GT$; its unconstrained maximiser $G^{-1}\alpha = (1, 0)$ lies in the box (on its boundary, where both partial derivatives $\alpha - GT$ vanish), so $T^{\mathrm{W}} = (1, 0)$ is the maximiser; it is unique because $G \succ 0$. The Nash potential of Lemma 3 is $\langle \alpha, T\rangle - \frac{1}{2}T^\top(G + \frac{1}{n}I)T$ (each type moves along a unit vector, so $\|L_\theta T_\theta\|^2 = T_\theta^2$); its unconstrained maximiser solves $(G + \frac{1}{n}I)T = \alpha$, and direct substitution (with $c = 1 - 1/n$) shows that T^{N} above solves it; it lies in the interior of the box, so it is the Nash equilibrium. Then $\Delta A = J(T^{\mathrm{N}} - T^{\mathrm{W}})$ with $T^{\mathrm{N}} - T^{\mathrm{W}} = -\big(\frac{1}{4} + \frac{1}{4n}, -\frac{1}{4} + \frac{1}{4n}\big)$, and $\|\Delta A\|^2 = \Delta T^\top G \Delta T$ evaluates (using $\sin^2\varphi_n = 1 - c^2$) to the stated expression, which is at least $1/(8n)$. With the Wardrop selection in which every A carrier delivers $1/n$ and every B carrier nothing, $\sum_i \|a_i^{\mathrm{W}}\|^2 = 1/n$. Per-carrier deliveries are at most $1/n$ by construction. □

The mechanism is visible in the eigenvectors of G: when two types move along nearly parallel directions (φ small), their *difference* direction $e_1 - v$ is almost free for the mean-field market, so a small difference between the types' own surcharges (Ma_i^{N}, of size $1/n$) moves the type totals by $O(1/(n\lambda_{\min}(G)))$ in opposite directions, and the aggregate by $O(1/(n\lambda_{\min}(G)^{1/2}))$. For a *fixed* angle φ the same computation gives $n\|\Delta A\|_2 \to 1/\sin\varphi$ as $n \to \infty$: the rate is $O(1/n)$, but with a constant that grows without bound as the types become collinear (Proposition 9 below states this in general). Proposition 5 answers the question of the memo negatively: an instance-free constant for the cross-zone case does not exist.

2.2 Exact $O(\bar{s})$ bounds under structure

Theorem 6 (Zone-separable carriers: constant one)**.** *Suppose every carrier's problem separates by coordinate: $Y_i = \prod_k Y_{ik}$, $\pi_i = \sum_k \pi_{ik}$, and a_{ik} depends on y_{ik} only, so that carrier i chooses a number of deliveries q_{ik} in a compact set $S_{ik} \subset \mathbb{R}_{\geq 0}$ (an interval for the convexified game, a finite set for lumpy plans) with concave (respectively arbitrary) profit π_{ik}. Then for every coordinate k and every charge u:*

(a) (convexified game) $0 \leq Q_k^{\mathrm{W}} - Q_k^{\mathrm{N}} \leq \max_i q_{ik}^{\mathrm{N}}$;

(b) (lumpy plans) every pure Nash equilibrium satisfies $0 \leq Q_k^{\mathrm{W}} - Q_k^{\mathrm{N}} \leq \max_i \max S_{ik}$, where Q^{W} is the Wardrop aggregate of the mixed extension.

In (a) equality holds when the N carriers are identical with linear profit and an interior equilibrium: then $Q_k^{\mathrm{N}} = \frac{N}{N+1} Q_k^{\mathrm{W}}$ and $Q_k^{\mathrm{W}} - Q_k^{\mathrm{N}} = Q_k^{\mathrm{W}}/(N+1) = q_{ik}^{\mathrm{N}}$.

Proof. Separability decouples the coordinates, so fix k, drop it from the notation and write $m = m_k$, $\pi_i = \pi_{ik}$, $Q = Q^{\mathrm{N}}$ and $q_i = q_i^{\mathrm{N}}$; the charge u_k is absorbed into π_i. Let $\hat{s}_i = q_i$ in case (a) and $\hat{s}_i = \max S_i$ in case (b). We first show two revealed-preference facts about the Nash choice q_i:

(i) for all $q' \in S_i$ with $q' < q_i$: $\pi_i(q_i) - mQq_i \geq \pi_i(q') - mQq'$;

(ii) for all $q' \in S_i$ with $q' > q_i$: $\pi_i(q_i) - m(Q + \hat{s}_i)q_i \geq \pi_i(q') - m(Q + \hat{s}_i)q'$.

In case (a), Lemma 2(c) says that q_i maximises $\pi_i(q) - m(Q+q_i)q$ over S_i, which implies (ii) directly and (i) because $m(Q+q_i)(q_i - q') \geq mQ(q_i - q')$ for $q' < q_i$. In case (b), the Nash condition against q' reads $\pi_i(q_i) - \pi_i(q') \geq m[(q_i - q')Q_{-i} + q_i^2 - q'^2] = m(q_i - q')(Q_{-i} + q_i + q')$ with $Q_{-i} = Q - q_i$; for $q' < q_i$ the right side is at least $m(q_i - q')Q$, and for $q' > q_i$ it is at least $-m(q' - q_i)(Q + \hat{s}_i)$ because $q' \leq \hat{s}_i$.

Now let $p \geq 0$ and let $\mathrm{BR}_i(p) = \arg\max_{q \in S_i} \pi_i(q) - pq$; in case (b) a Wardrop carrier mixes over $\mathrm{BR}_i(p)$, in case (a) it chooses a point of the interval $\mathrm{BR}_i(p)$; either way its delivery lies in $\operatorname{conv} \mathrm{BR}_i(p)$ and $Q^{\mathrm{W}} = \sum_i \bar{q}_i^{\mathrm{W}}$ with $\bar{q}_i^{\mathrm{W}} \in \operatorname{conv} \mathrm{BR}_i(mQ^{\mathrm{W}})$.

Upper bound $Q \leq Q^{\mathrm{W}}$. Suppose $Q > Q^{\mathrm{W}}$, so $p := mQ^{\mathrm{W}} < mQ$. For $q' < q_i$, (i) gives $\pi_i(q_i) - pq_i - (\pi_i(q') - pq') \geq (mQ - p)(q_i - q') > 0$, so no $q' < q_i$ is a best response at p, and $\bar{q}_i^{\mathrm{W}} \geq q_i$ for every i. Summing, $Q^{\mathrm{W}} \geq Q$, a contradiction.

Lower bound $Q \geq Q^{\mathrm{W}} - \hat{s}$, $\hat{s} = \max_i \hat{s}_i$. Suppose $Q < Q^{\mathrm{W}} - \hat{s}$, so $p := mQ^{\mathrm{W}} > m(Q + \hat{s}) \geq m(Q + \hat{s}_i)$. For $q' > q_i$, (ii) gives $\pi_i(q_i) - pq_i - (\pi_i(q') - pq') \geq (p - m(Q + \hat{s}_i))(q' - q_i) > 0$, so $\bar{q}_i^{\mathrm{W}} \leq q_i$ for all i and $Q^{\mathrm{W}} \leq Q$, a contradiction.

Equality case: with profit αq and an interior solution, the first-order conditions are $\alpha = mQ^{\mathrm{W}}$ and $\alpha = mQ^{\mathrm{N}} + mq$ with $Q^{\mathrm{N}} = Nq$, so $q = \alpha/(m(N+1))$ and $Q^{\mathrm{W}} - Q^{\mathrm{N}} = \alpha/m - N\alpha/(m(N+1)) = q$. □

The theorem needs no symmetry, no smoothness and no bound on the number of carriers, and the constant is attained. Its scope is a carrier whose plans in different zone-hours do not compete for the same fleet. A carrier whose fleet budget is shared across zones, or whose tours visit several zones, is not separable; Proposition 5 shows that for such carriers no analogue with an instance-free constant holds.

Proposition 7 (Equal own surcharges)**.** *If all carriers have the same Nash contribution $\bar{a}$ (in particular, if they are identical: Lemma 3), then $\|A^{\mathrm{N}} - A^{\mathrm{W}}\|_m \leq \|\bar{a}\|_m = \|A^{\mathrm{N}}\|_m/N$, for any coupling across coordinates.*

Proof. In the first inequality of Theorem 4(a), $\sum_i \langle \bar{a}, M(a_i^{\mathrm{W}} - \bar{a})\rangle = \langle \bar{a}, M(A^{\mathrm{W}} - A^{\mathrm{N}})\rangle \leq \|\bar{a}\|_m \|\Delta A\|_m$. □

Theorem 8 (Replicas: $O(1/n)$ with an instance-dependent constant)**.** *Consider a replica game with base types $\theta \in \Theta$, polyhedral $\mathcal{Y}_\theta$ and linear profits, and let $n = \min_\theta n_\theta$. There is a constant C, depending on the base instance (and on u) but not on (n_θ), such that $\|A^{\mathrm{N}} - A^{\mathrm{W}}\|_m \leq C/n$ for all $n \geq 1$.*

Proof. For $\sigma = (\sigma_\theta)_\theta \in \mathbb{R}^{d \times \Theta}$ let $S(\sigma)$ be the Wardrop aggregate of the base types (one carrier per type) when type θ pays the extra per-delivery charge σ_θ; it is unique by Lemma 2(b), and $S(0) = A^{\mathrm{W}}$ for every replica game (Lemma 3). By Lemmas 3 and 2(c), the Nash aggregate of the replica game is $S(\sigma^n)$ with $\sigma_\theta^n = ML_\theta Y_\theta^{\mathrm{N}}/n_\theta$, so $\|\sigma^n\| \leq c_1/n$ with $c_1 = \max_\theta \max_{Y \in \mathcal{Y}_\theta} \|ML_\theta Y\|$. The Wardrop

equilibria under σ are the solutions of the Karush–Kuhn–Tucker system of a convex quadratic program with polyhedral constraints in which σ enters linearly; its solution set, as a multifunction of σ, has a graph that is a finite union of polyhedra. By Robinson [8] such a multifunction is locally upper Lipschitz: there are $\eta, C_0 > 0$ with $S(\sigma) \in S(0) + C_0\|\sigma\|\mathbb{B}$ whenever $\|\sigma\| \le \eta$. Hence $\|\Delta A\| \le C_0 c_1/n$ for $n \ge c_1/\eta$, and $\|\Delta A\| \le D \le D(c_1/\eta)/n$ otherwise, where D is the diameter of the set of feasible aggregates. □

The constant of Theorem 8 is not explicit. In the smooth interior case it is:

Proposition 9 (Affine interior replicas: explicit constant)**.** *Let base type θ choose t_θ in an open set, deliver $L_\theta t_\theta$ and earn $\langle c_\theta, t_\theta\rangle - \frac{1}{2}t_\theta^\top K_\theta t_\theta$ with $K_\theta \succeq 0$ (its n_θ-split carriers earn $\langle c_\theta, t\rangle - \frac{n_\theta}{2}t^\top K_\theta t$). Let $L = [L_\theta]_\theta$, $G = L^\top M L$, $K = \operatorname{diag}(K_\theta)$, $D_n = \operatorname{diag}(L_\theta^\top M L_\theta/n_\theta)$, and suppose $G + K \succ 0$ and that both equilibria are interior. Then*

$$\|A^{\mathrm{N}} - A^{\mathrm{W}}\|_m = \|G^{1/2}(G+K+D_n)^{-1}D_n T^{\mathrm{W}}\|_2 \le \|D_n T^{\mathrm{W}}\|_{(G+K)^{-1}} \le \lambda_{\min}(G+K)^{-1/2}\Big(\sum_\theta \|L_\theta^\top M a_\theta^{\mathrm{W}}\|^2\Big)^{1/2},$$

with $a_\theta^{\mathrm{W}} = L_\theta T_\theta^{\mathrm{W}}/n_\theta$ the per-carrier Wardrop delivery of type θ.

Proof. The first-order conditions are $(G+K)T^{\mathrm{W}} = c - L^\top u$ and $(G+K+D_n)T^{\mathrm{N}} = c - L^\top u$, so $\Delta T = -(G+K+D_n)^{-1}D_n T^{\mathrm{W}}$ and $\|\Delta A\|_m^2 = \Delta T^\top G \Delta T$. With $S = G + K + D_n \succeq G + K \succ 0$, $S^{-1}GS^{-1} \preceq S^{-1} \preceq (G+K)^{-1}$, which gives the first inequality; the second is $\|x\|_{(G+K)^{-1}} \le \lambda_{\min}(G+K)^{-1/2}\|x\|_2$ and $(D_n T^{\mathrm{W}})_\theta = L_\theta^\top M a_\theta^{\mathrm{W}}$. □

When the profit of each type is linear ($K = 0$), the constant is $\lambda_{\min}(G)^{-1/2}$: the gap is $O(1/n)$ exactly when the types' delivery directions are linearly independent in the M-geometry, and the constant degrades as they approach collinearity, as in Proposition 5 where $\lambda_{\min}(G) = 1 - \cos\varphi$. With strictly concave individual profits ($K \succ 0$) the constant is at most $\lambda_{\min}(K)^{-1/2}$; this is the regime of Paccagnan et al. [7, Thm. 1(1)–(2)], whose $O(1/N)$ bounds use strong monotonicity of the Nash or Wardrop operator in the individual strategies. Routing plans with linear costs and fleet budgets do not provide such monotonicity, which is why Theorem 8 has to rely on polyhedrality.

Summary of the rate question With $\bar{s}$ the largest share of one carrier at one coordinate: the gap is at most $\frac{1}{2}(\bar{s}\langle m, A^{\mathrm{W}}\rangle)^{1/2}$ always (Theorem 4), and this order is attained (Proposition 5); it is at most $\bar{s}$ per coordinate with constant one for zone-separable carriers (Theorem 6) and at most $\|A^{\mathrm{N}}\|_m/N$ for identical carriers (Proposition 7); for replicas of a fixed instance it is $O(1/n)$ with a constant that depends on the conditioning of the instance (Theorem 8, Proposition 9).

3 The city's charge in the limit

The city posts a per-delivery charge $u \in \mathbb{R}^d$ (by Omega Proposition "one instrument per zone" it can be realised as a uniform per-delivery charge plus non-negative congestion charges per vehicle-hour). Let $Q^{\mathrm{N}}(u)$ and $Q^{\mathrm{W}}(u)$ be the Nash and Wardrop aggregates under u (unique by Lemma 2). The *observation-only rule* of the Omega paper, in its form without the market-power subsidy, posts

$$u^{t+1} = (1-\theta)u^t + \theta\,\kappa \odot Q(u^t), \qquad \theta \in (0,1],$$

using only the observed aggregate: the corrective charge at the observed deliveries. In the mean-field limit the market-power subsidy of the Omega rule has no object, and this is the natural rule.

Lemma 10 (Responses are nonexpansive)**.** *For $X \in \{\mathrm{N}, \mathrm{W}\}$ and charges u, u': $\|Q^X(u) - Q^X(u')\|_m^2 \le \langle u' - u, Q^X(u) - Q^X(u')\rangle$, hence $\|Q^X(u) - Q^X(u')\|_m \le \|u - u'\|_{m^{-1}}$.*

Proof. For $X = \mathrm{W}$ write the two optimality inequalities of Theorem 4(a) with the two equilibria $y = y^{\mathrm{W}}(u)$, $y' = y^{\mathrm{W}}(u')$: $\pi_i(y_i') - \langle a_i', MA + u\rangle \le \pi_i(y_i) - \langle a_i, MA + u\rangle$ and the symmetric one. Adding and summing over i gives $\|A - A'\|_m^2 + \langle A - A', u - u'\rangle \le 0$. For $X = \mathrm{N}$ the surcharges Ma_i, Ma_i' of Lemma 2(c) add $\sum_i \|a_i - a_i'\|_m^2 \ge 0$ to the left side. The second inequality follows by Cauchy–Schwarz in the dual pair $(\|\cdot\|_m, \|\cdot\|_{m^{-1}})$. □

Proposition 11 (Contraction and fixed point)**.** *Let $T_X(u) = \kappa \odot Q^X(u)$ for $X \in \{\mathrm{N}, \mathrm{W}\}$.*

(a) $\|T_X(u) - T_X(u')\|_{\kappa^{-1}} \le \varrho \|u - u'\|_{\kappa^{-1}}$ with $\varrho = \max_k \kappa_k/m_k < 1$; the plain rule ($\theta = 1$) converges linearly with ratio ϱ.

(b) The unique fixed point is $u_X^* = \kappa \odot \tilde{Q}_X$, where $\tilde{Q}_X$ is the aggregate of any maximiser of $\tilde{\Phi}_X$, defined as Φ_X with $u = 0$ and the coupling $\frac{1}{2}\|A\|_m^2$ replaced by $\frac{1}{2}\|A\|_{m+\kappa}^2$ (the own terms of Φ_{N} keep the weight m).

(c) $\tilde{\Phi}_{\mathrm{W}} = \mathrm{SW}$, so $u_{\mathrm{W}}^* = \kappa \odot Q^{\mathrm{opt}}$ is the marginal-cost (Pigou) charge and the Wardrop equilibrium under u_{W}^* is socially optimal.

Proof. (a) $\|\kappa \odot \Delta Q\|_{\kappa^{-1}}^2 = \sum_k \kappa_k \Delta Q_k^2 \le \varrho \|\Delta Q\|_m^2 \le \varrho \|\Delta u\|_{m^{-1}}^2 \le \varrho^2 \|\Delta u\|_{\kappa^{-1}}^2$, using $\kappa_k \le \varrho m_k$ twice and Lemma 10. (b) Let $\tilde{y}$ maximise $\tilde{\Phi}_X$ and $\tilde{Q}$ its aggregate. Lemma 1, applied to the convex term $\frac{1}{2}\|A\|_\kappa^2$, shows that $\tilde{y}$ maximises $\tilde{\Phi}_X + \frac{1}{2}\|A\|_\kappa^2 - \langle \kappa \odot \tilde{Q}, A\rangle$, which is Φ_X with $u = \kappa \odot \tilde{Q}$. By Lemma 2(a), $\tilde{y}$ is an X-equilibrium under that charge, so $\kappa \odot \tilde{Q}$ is a fixed point; it is unique by (a). (c) $\frac{1}{2}\|A\|_{m+\kappa}^2 = \frac{1}{2}\|Q\|_{b+2\kappa}^2$ is the quadratic part of (2). □

Part (a) is the contraction theorem of the Omega paper ("the response is a proximal map"), restated for the time-expanded game and for both behaviours; part (c) is the classical marginal-cost toll [1], here reached by observation alone. Part (b) is what makes the finite-N analysis exact:

Theorem 12 (Charge convergence)**.** *Let $\tilde{Q}_{\mathrm{N}}, \tilde{Q}_{\mathrm{W}}$ be the Nash and Wardrop aggregates of the auxiliary game of Proposition 11(b) (coupling $m + \kappa$, own weight m). Then*

$$u_{\mathrm{N}}^* - u_{\mathrm{W}}^* = \kappa \odot (\tilde{Q}_{\mathrm{N}} - \tilde{Q}_{\mathrm{W}}), \qquad \|u_{\mathrm{N}}^* - u_{\mathrm{W}}^*\|_{\kappa^{-1}} = \|\tilde{Q}_{\mathrm{N}} - \tilde{Q}_{\mathrm{W}}\|_\kappa,$$

and consequently

(a) in general, $\|u_{\mathrm{N}}^* - u_{\mathrm{W}}^*\|_{\kappa^{-1}} \le \frac{1}{2}\tilde{\varrho}^{1/2}\big(\sum_i \|\tilde{a}_i^{\mathrm{W}}\|_m^2\big)^{1/2} \le \frac{1}{2}\big(\tilde{\varrho}\,\bar{s}\,\langle m, \tilde{Q}_{\mathrm{W}}\rangle\big)^{1/2}$ with $\tilde{\varrho} = \max_k \kappa_k/(m_k + \kappa_k)$;

(b) for zone-separable carriers, $0 \le u_{\mathrm{W},k}^* - u_{\mathrm{N},k}^* \le \kappa_k \max_i \tilde{q}_{ik}^{\mathrm{N}} \le \kappa_k \bar{s}$ for every k;

(c) for replicas (polyhedral types, linear profits), $\|u_{\mathrm{N}}^* - u_{\mathrm{W}}^*\| = O(1/n)$ with an instance-dependent constant.

Moreover, for any charge-to-charge perturbation: if T is a ϱ-contraction with fixed point u^* and u' is a fixed point of a map T' with $\|T'u' - Tu'\| \le \varepsilon$, then $\|u' - u^*\| \le \varepsilon/(1 - \varrho)$; with $T = T_{\mathrm{W}}$, $T' = T_{\mathrm{N}}$ this gives the memo's form $\|u_{\mathrm{N}}^* - u_{\mathrm{W}}^*\|_{\kappa^{-1}} \le \|Q^{\mathrm{N}}(u_{\mathrm{N}}^*) - Q^{\mathrm{W}}(u_{\mathrm{N}}^*)\|_\kappa/(1 - \varrho)$.

Proof. The identity is Proposition 11(b). The auxiliary game is a game of the form of Section 1 with coupling $m' = m + \kappa$ except that each carrier's own term carries m instead of m'; Lemma 2(c) then gives own surcharges Ma_i instead of $M'a_i$, and the proofs of Theorems 4, 6 and 8 go through verbatim: in Theorem 4(a) one obtains $\|\Delta\tilde{Q}\|^2_{m'} \le \sum_i \langle \tilde{a}_i^{\mathrm{N}}, M(\tilde{a}_i^{\mathrm{W}} - \tilde{a}_i^{\mathrm{N}})\rangle \le \frac{1}{4}\sum_i \|\tilde{a}_i^{\mathrm{W}}\|^2_m$, and $\|x\|^2_\kappa \le \tilde{\varrho}\|x\|^2_{m'}$ gives (a); in Theorem 6 the surcharge $m_k q_i \le m'_k q_i$ only strengthens (ii), so $0 \le \tilde{Q}_{\mathrm{W},k} - \tilde{Q}_{\mathrm{N},k} \le \max_i \tilde{q}_{ik}^{\mathrm{N}}$, which is (b) after multiplying by κ_k; (c) is Theorem 8. The perturbation statement: $\|u' - u^*\| = \|T'u' - Tu^*\| \le \|T'u' - Tu'\| + \|Tu' - Tu^*\| \le \varepsilon + \varrho\|u' - u^*\|$. □

So the finite-market fixed point lies *below* the Pigou charge in every coordinate when carriers are zone-separable, and the shortfall is at most one carrier's deliveries times the congestion slope. The rate of convergence of the charge is the rate of convergence of the aggregate in the auxiliary game; the factor $1/(1-\varrho)$ of the perturbation argument is not needed.

Theorem 13 (Welfare)**.** *Let y^{opt} be a social optimum with aggregate Q^{opt} and $\mathrm{SW}^{\mathrm{opt}} = \mathrm{SW}(y^{\mathrm{opt}})$.*

(a) For every charge v and every Wardrop equilibrium y under v with aggregate Q,

$$\mathrm{SW}^{\mathrm{opt}} - \mathrm{SW}(y) \le \langle v - \kappa \odot Q^{\mathrm{opt}}, Q^{\mathrm{opt}} - Q\rangle - \tfrac{1}{2}\|Q - Q^{\mathrm{opt}}\|^2_b \le \tfrac{1}{2}\|v - \kappa \odot Q^{\mathrm{opt}}\|^2_{b^{-1}}.$$

(b) (Uniform $O(\bar{s})$ efficiency of the observation-only charge.) The Nash equilibrium under the finite-market fixed point u^*_{N} satisfies

$$0 \le \mathrm{SW}^{\mathrm{opt}} - \mathrm{SW}(y^{\mathrm{N}}(u^*_{\mathrm{N}})) \le \tfrac{1}{2}\sum_i \|a_i^{\mathrm{opt}}\|^2_m - \tfrac{1}{2}\sum_i \|a_i^{\mathrm{N}}\|^2_m \le \tfrac{1}{2}\bar{s}\,\langle m, Q^{\mathrm{opt}}\rangle.$$

(c) For every charge u, $\big|\mathrm{SW}(y^{\mathrm{N}}(u)) - \mathrm{SW}(y^{\mathrm{W}}(u))\big| \le \frac{1}{2}\sum_i \|a_i^{\mathrm{W}}(u)\|^2_m + \|Q^{\mathrm{N}}(u) - Q^{\mathrm{W}}(u)\|_\kappa \, \|\bar{Q}(u) - \kappa^{-1} \odot u\|_\kappa$, where $\bar{Q}(u)$ is the midpoint of the two aggregates.

Proof. Write $P^v(y) = \sum_i \pi_i - \frac{1}{2}\|Q\|^2_m - \langle v, Q\rangle$, so $\mathrm{SW} = P^v + \langle v, Q\rangle - \frac{1}{2}\|Q\|^2_\kappa$. (a) y maximises P^v, which is the sum of a concave function and $-\frac{1}{2}\|Q\|^2_m$; by Lemma 1, y also maximises $\sum_i \pi_i - \langle v, Q\rangle - \langle MQ, \cdot\rangle$ evaluated at the aggregate, and the exact expansion of the quadratic term then gives $P^v(y) - P^v(y^{\mathrm{opt}}) \ge \frac{1}{2}\|Q - Q^{\mathrm{opt}}\|^2_m$. Hence $\mathrm{SW}^{\mathrm{opt}} - \mathrm{SW}(y) \le -\frac{1}{2}\|\Delta Q\|^2_m + \langle v, Q^{\mathrm{opt}} - Q\rangle - \frac{1}{2}\|Q^{\mathrm{opt}}\|^2_\kappa + \frac{1}{2}\|Q\|^2_\kappa$ with $\Delta Q = Q - Q^{\mathrm{opt}}$, and $\frac{1}{2}\|Q\|^2_\kappa - \frac{1}{2}\|Q^{\mathrm{opt}}\|^2_\kappa = \langle \kappa \odot Q^{\mathrm{opt}}, \Delta Q\rangle + \frac{1}{2}\|\Delta Q\|^2_\kappa$; with $m - \kappa = b$ this is the first inequality, and maximising over ΔQ gives the second. (b) By Proposition 11(b), $y^{\mathrm{N}}(u^*_{\mathrm{N}})$ maximises $\tilde{\Phi}_{\mathrm{N}} = \mathrm{SW} - \frac{1}{2}\sum_i \|a_i\|^2_m$; compare with y^{opt}. The last inequality uses $0 \le a_{ik}^{\mathrm{opt}} \le \bar{s}$. (c) $\mathrm{SW} = P^u - \frac{1}{2}\|Q - \kappa^{-1}\odot u\|^2_\kappa + \frac{1}{2}\|u\|^2_{\kappa^{-1}}$. Lemma 2 gives $0 \le P^u(y^{\mathrm{W}}) - P^u(y^{\mathrm{N}}) \le \frac{1}{2}\sum_i \|a_i^{\mathrm{W}}\|^2_m - \frac{1}{2}\sum_i \|a_i^{\mathrm{N}}\|^2_m$ (because y^{N} maximises $P^u - \frac{1}{2}\sum \|a_i\|^2_m$), and the difference of the two quadratic terms is $\langle Q^{\mathrm{N}} - Q^{\mathrm{W}}, \bar{Q} - \kappa^{-1} \odot u\rangle_\kappa$. □

Part (b) is uniform: unlike the aggregates (Proposition 5), the welfare of the observation-only charge converges at rate $O(\bar{s})$ with constant $\frac{1}{2}\langle m, Q^{\mathrm{opt}}\rangle$ for every instance. The reason is that welfare is second order in the aggregate error near the optimum. Part (b) does not say that the charge helps when N is small: with few carriers the Cournot shortfall dominates and a charge that only adds the congestion cost can lower welfare (Section). For a fixed small N the Omega rule with its market-power subsidy is the appropriate one.

Corollary 14 (Price of anarchy and leader gain)**.** *Let $\mathrm{PoA}_X = \mathrm{SW}^{\mathrm{opt}}/\mathrm{SW}(y^X(0))$ (welfare positive) and $G_X = \mathrm{SW}(y^X(u^*_X)) - \mathrm{SW}(y^X(0))$ the gain of the observation-only charge. Then $|\mathrm{PoA}_{\mathrm{N}} - \mathrm{PoA}_{\mathrm{W}}| \le \mathrm{PoA}_{\mathrm{N}}\mathrm{PoA}_{\mathrm{W}}\,|\mathrm{SW}(y^{\mathrm{N}}(0)) - \mathrm{SW}(y^{\mathrm{W}}(0))|/\mathrm{SW}^{\mathrm{opt}}$ and $|G_{\mathrm{N}} - G_{\mathrm{W}}| \le \big[\mathrm{SW}^{\mathrm{opt}} - \mathrm{SW}(y^{\mathrm{N}}(u^*_{\mathrm{N}}))\big] + |\mathrm{SW}(y^{\mathrm{N}}(0)) - \mathrm{SW}(y^{\mathrm{W}}(0))|$; by Theorems 13 and 4–8 both tend to zero at rate $O(\bar{s}) + O(\|Q^{\mathrm{N}}(0) - Q^{\mathrm{W}}(0)\|_\kappa)$, that is $O(\bar{s})$ for separable carriers and replicas and $O(\bar{s}^{1/2})$ in the worst case.*

Proof. Direct from the definitions, $\mathrm{SW}(y^{\mathrm{W}}(u^*_{\mathrm{W}})) = \mathrm{SW}^{\mathrm{opt}}$ (Proposition 11(c)) and Theorem 13(b),(c) at $u = 0$. □

4 Cautious carriers in the limit

Following decision (4) of the memo, only two pieces of the imprecise-probability layer of the Omega paper are kept: the Γ-maximin surcharge and the scaling of the belief width needed for credal absorption. Fix weights $w \in \mathbb{R}^d_{>0}$ and a width $\delta \geq 0$; a carrier believes that the mean rival aggregate lies in the box $\{A' : |A'_k - A_{-i,k}| \leq \delta w_k\}$.

Proposition 15 (Γ-maximin carriers)**.** *(a) (Omega, Lemma "Γ-maximin surcharge".) Because deliveries are nonnegative and the coupling is affine, the lower expectation of every plan is attained at the upper corner $A_{-i} + \delta w$. Hence the Γ-maximin equilibria, finite or mean-field, are the precise equilibria under the charge $u + \delta\, m \odot w$: $Q^{X,\delta}(u) = Q^X(u + \delta m \odot w)$.*

(b) $\|Q^{X,\delta}(u) - Q^X(u)\|_m \leq \delta \|w\|_m$; for zone-separable carriers, $0 \leq Q^X_k(u) - Q^{X,\delta}_k(u) \leq \delta w_k$ for $X = \mathrm{W}$.

(c) The observation-only rule under Γ-maximin carriers, $u \mapsto \kappa \odot Q^{X,\delta}(u)$, contracts with the same ratio ϱ (it needs no knowledge of δ); its fixed point u^*_δ satisfies $\|u^*_\delta - u^*_X\|_{\kappa^{-1}} \leq \varrho^{1/2}\delta\|w\|_m/(1-\varrho)$.

(d) In the mean-field game, the equilibrium under u^*_δ loses at most $\frac{1}{2}\delta^2\|m \odot w\|^2_{(m+\kappa)^{-1}}$ of welfare. A city that knows δw and posts $u^*_\delta - \delta m \odot w$ recovers the optimum.

Proof. (a) is the Omega lemma; in the time-expanded model the coordinates are zone-hours, and nothing else changes. (b) Lemma 10 with $u' = u + \delta m \odot w$: $\|\delta m \odot w\|_{m^{-1}} = \delta\|w\|_m$. In the separable case the proof of Theorem 6 with a common price increase $\delta m_k w_k$ in coordinate k (in place of the private surcharges) gives the coordinatewise statement: the Wardrop aggregate falls, and by at most the price increase divided by m_k. (c) The map is $T_X(\cdot + \delta m \odot w)$, a ϱ-contraction in $\|\cdot\|_{\kappa^{-1}}$ (Proposition 11). Its distance to T_X at any u is $\|Q^X(u + \delta m \odot w) - Q^X(u)\|_\kappa \leq \varrho^{1/2}\|\cdot\|_m \leq \varrho^{1/2}\delta\|w\|_m$; apply the perturbation statement of Theorem 12. (d) The carriers face $v = u^*_\delta + \delta m \odot w$ and $u^*_\delta = \kappa \odot Q$ at the fixed point, so in Theorem 13(a) $v - \kappa \odot Q^{\mathrm{opt}} = \kappa \odot \Delta Q + \delta m \odot w$ with $\Delta Q = Q - Q^{\mathrm{opt}}$, and the bound becomes $-\langle \delta m \odot w, \Delta Q\rangle - \frac{1}{2}\|\Delta Q\|^2_{b+2\kappa} \leq \frac{1}{2}\delta^2\|m \odot w\|^2_{(b+2\kappa)^{-1}}$, with $b + 2\kappa = m + \kappa$. □

So cautious carriers behave like carriers facing a tax of $\delta m_k w_k$ per delivery; the observation-only city cannot tell this tax apart from cost, and over-charges by about that amount. The welfare cost is second order in δ and of Harberger form.

Corollary 16 (Belief width needed for absorption)**.** *Let $\sigma_{\max}$ be the largest range of one carrier's deliveries at one coordinate in units of w. The uniform credal-absorption bound of the Omega paper, $\delta^E_0 \leq \sigma_{\max} \min\{N - 1, d + 1, 1 + \sqrt{d \ln(2d)/2}\}$, is at most $(d+1)\bar{s}/\min_k w_k$. For n-splits of a fixed instance $\bar{s} \propto 1/n$, so the width at which a rounded pure profile becomes an E-admissible equilibrium is $O(d/n)$ and vanishes in the limit; the N-free terms bind exactly when $N > d + 1$.*

Proof. $\sigma_{\max} \leq \bar{s}/\min_k w_k$ and $\min\{\cdot\} \leq d + 1$. □

With $d = ZH$ coordinates the terms $d + 1$ and $1 + \sqrt{d\ln(2d)/2}$ become relevant only for very many carriers (for Cairo with 179 active zone-hours in the experiments, $N > 180$), which is the regime of gig couriers and small operators. The corollary is algebra on the Omega theorem; its content is the regime statement.

Proposition 17 (Time-of-day charges can move the worst case of travel times)**.** *In the Omega routing layer the cost of a route is nondecreasing in every arc travel time, and the upper prevision over a probability box with unknown dependence is attained at the upper (slowest) marginals. If instead a period-dependent charge is paid at the arrival time, monotonicity can fail: with one arc whose travel time T has a p-box containing all laws on $[0,2]$, and cost $T+U\cdot\mathbf{1}\{T<1\}$ with $U>2$, the upper prevision equals $\sup_{T\in[0,2]}(T+U\mathbf{1}\{T<1\})=1+U$ (approached as $T\uparrow 1$), whereas the slowest law ($T=2$) costs 2.*

Proof. The p-box contains every Dirac law on $[0,2]$, so the upper prevision is the supremum of the cost; on $[0,1)$ the cost is $T+U<1+U$ and tends to $1+U$, on $[1,2]$ it is at most $2<1+U$. □

When charges increase over the day (or are constant), the arrival-time cost is again nondecreasing in travel time and the Omega corner property is restored; with decreasing charges, the robust route cost must be computed over the whole box. In the experiments below travel times are deterministic, so this proposition limits the transfer of the Omega routing layer to the time-expanded model rather than affecting the computations.

5 Vehicle-count congestion

Under D2 the congestion variable counts vehicles. Let mean-field types $\theta\in[\Theta]$ choose exposures $\tau_\theta\in\mathcal{T}_\theta\subset\mathbb{R}^Z_{\geq 0}$ (vehicle-hours per zone; compact convex) with concave value V_θ, and let a type-θ vehicle dwell $h_{\theta z}>0$ hours in zone z, so that it contributes $\tau_{\theta z}/h_{\theta z}$ vehicles. The congestion variable is $X_z=\sum_\theta \tau_{\theta z}/h_{\theta z}$ and a vehicle-hour in z costs g_zX_z, $g_z>0$. A mean-field equilibrium is a profile in which each τ_θ maximises $V_\theta(\tau)-\langle\tau, g\odot X\rangle$ at the realised X. The coupling operator $F_\theta(\tau)=g\odot X(\tau)$ has Jacobian blocks $\partial F_\theta/\partial\tau_{\theta'}=\mathrm{diag}_z(g_z/h_{\theta' z})$.

Proposition 18 (Existence and the rank-one condition)**.** *(a) An equilibrium exists.*

(b) If $h_{\theta z}=\eta_\theta\zeta_z$ (rank one), the game is a weighted potential game with weights $1/\eta_\theta$ and concave potential $\Psi(\tau)=\sum_\theta \eta_\theta^{-1}V_\theta(\tau_\theta)-\frac{1}{2}\sum_z g_z\zeta_zX_z^2$; equilibria are the maximisers of Ψ, and X is unique when the V_θ are concave.

(c) Positive weights w_θ with $w_\theta\,\partial F_\theta/\partial\tau_{\theta'}$ symmetric for all θ,θ' exist iff $h_{\theta z}/h_{\theta' z}$ does not depend on z, i.e. iff h is rank one.

(d) With two types in one zone and $h_1\neq h_2$, the symmetric part of the coupling matrix $g\left(\begin{smallmatrix}1/h_1 & 1/h_2\\ 1/h_1 & 1/h_2\end{smallmatrix}\right)$ has determinant $-g^2(1/h_1-1/h_2)^2/4<0$, so the unweighted operator is not monotone.

Proof. (a) Each type's best-response correspondence at a given X is nonempty, convex-valued and upper hemicontinuous (Berge), and X is continuous in τ; Kakutani's theorem on $\prod_\theta\mathcal{T}_\theta$ gives a fixed point. (b) $\partial\Psi/\partial\tau_{\theta z}$ of the coupling part is $-g_z\zeta_zX_z/(\eta_\theta\zeta_z)=-\eta_\theta^{-1}g_zX_z$, which is η_θ^{-1} times the marginal coupling cost of type θ; Ψ is concave and the equivalence of equilibria and maximisers follows as in Lemma 2(a); X is a linear image on which Ψ is strictly concave. (c) Symmetry means $w_\theta g_z/h_{\theta' z}=w_{\theta'}g_z/h_{\theta z}$, i.e. $h_{\theta z}/h_{\theta' z}=w_{\theta'}/w_\theta$ for all z. (d) Direct computation. □

In the finite game, deciding whether a pure equilibrium exists under D2 is strongly NP-complete (Omega, Theorem "NP-completeness under D2"); in the convexified mean-field game an equilibrium always exists, so the limit removes the existence problem. What the limit does not remove is non-monotonicity: with dwell times that are not rank one there is no weighting under which the game is a potential game.

Conjecture 19. *For rank$(h) \geq 2$ there are instances with several isolated mean-field equilibria, and computing one is PPAD-hard in the number of types.*

Neither part has been checked; no reduction has been designed. The rank-one condition is plausible for a fleet of one vehicle class (dwell time proportional to a zone factor) and implausible for mixed fleets of vans and cargo bikes.

References

[1] Martin Beckmann, C. B. McGuire, and Christopher B. Winsten. *Studies in the Economics of Transportation*. Yale University Press, 1956.

[2] Dan Calderone and S. Shankar Sastry. Markov decision process routing games. In *Proceedings of the 8th International Conference on Cyber-Physical Systems (ICCPS)*, pages 273–279, 2017.

[3] Quang-Vinh Dang. City charges for competing delivery carriers: exact best responses under probability-box travel times and leader pricing without equilibrium search. Working paper, under review, 2026.

[4] Diogo A. Gomes, Joana Mohr, and Rafael R. Souza. Discrete time, finite state space mean field games. In *Dynamics, Games and Science I*. Springer, 2011.

[5] Olivier Guéant. Existence and uniqueness result for mean field games with congestion effect on graphs. *Applied Mathematics & Optimization*, 72(2):291–303, 2015.

[6] Dov Monderer and Lloyd S. Shapley. Potential games. *Games and Economic Behavior*, 14(1): 124–143, 1996.

[7] Dario Paccagnan, Basilio Gentile, Francesca Parise, Maryam Kamgarpour, and John Lygeros. Nash and Wardrop equilibria in aggregative games with coupling constraints. *IEEE Transactions on Automatic Control*, 64(4):1373–1388, 2019.

[8] Stephen M. Robinson. Some continuity properties of polyhedral multifunctions. *Mathematical Programming Study*, 14:206–214, 1981.

[9] Margaret E. Slade. What does an oligopoly maximize? *The Journal of Industrial Economics*, 42(1):45–61, 1994.